

Professional Ethics and Communication for Engineering

Class code



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Learning Outcomes

- To learn about the benefits of studying Engineering Ethics
- To learn how engineers manage the unknown
- To learn about Personal vs. Professional Ethics
- To learn about the types of Professional Ethics
- To learn about the types of Management Ethics

Example:

Software, civil, electrical, and mechanical engineers all face ethical issues.

Question:

- Why should engineers study ethics?

What is Ethics?



- ❖ Morality encompasses the first-order beliefs and practices about good and evil by which we guide our behaviour.
- ❖ Ethics is the second-order, reflective consideration of our moral beliefs and practices (Hinman, 2003; Baura, 2006)
- ❖ Meta-ethics: What is the meaning of ‘good’?
- ❖ Normative Ethics: It develops theories and models for how we should treat each other.
- ❖ Applied Ethics: It centers on the resolution of scientific ethical problems.

1. Meta-Ethics in Engineering

Meta-ethics asks about the **meaning of moral claims in engineering.**

Example:

Statement:

“An engineer must ensure public safety.”

Meta-ethical questions:

- What does **“must”** mean here — moral duty or legal duty?
- Is **“public safety is most important”** an objective truth or a professional agreement?
- Are engineering ethics universal or dependent on country and culture?

 Real-life context:

When different countries allow different safety limits (e.g., pollution level, radiation level), meta-ethics asks:

→ Are these limits morally right by nature, or socially decided?

2. Normative Ethics in Engineering

Normative ethics gives **general rules engineers should follow.**

Engineering Examples:

1. Safety Rule

“Engineers should prioritize public safety over cost and profit.”

2. Honesty Rule

“Engineers should report true test results and not falsify data.”

3. Professional Duty

“Engineers should not approve designs they know are unsafe.”

 Real-life context:

Professional codes (IEEE, ACM, NSPE) are based on **normative ethics.**

Example rule:

“Hold paramount the safety, health, and welfare of the public.”

This is a **normative ethical principle.**

3. Applied Ethics in Engineering

Applied ethics deals with **specific engineering cases**.

Example 1: Bridge Construction

An engineer discovers cracks in a newly built bridge design.

Question:

Should the engineer report the defect even if it delays the project and causes financial loss?

This is **applied ethics**.

Example 2: Software Engineering

A software engineer finds a bug that may leak user data.

Question:

Should the engineer inform management or hide it to meet the deadline?

This is **applied ethics**.

Example 3: Electrical Engineering

An engineer is asked to reduce safety components to lower production cost.

Question:

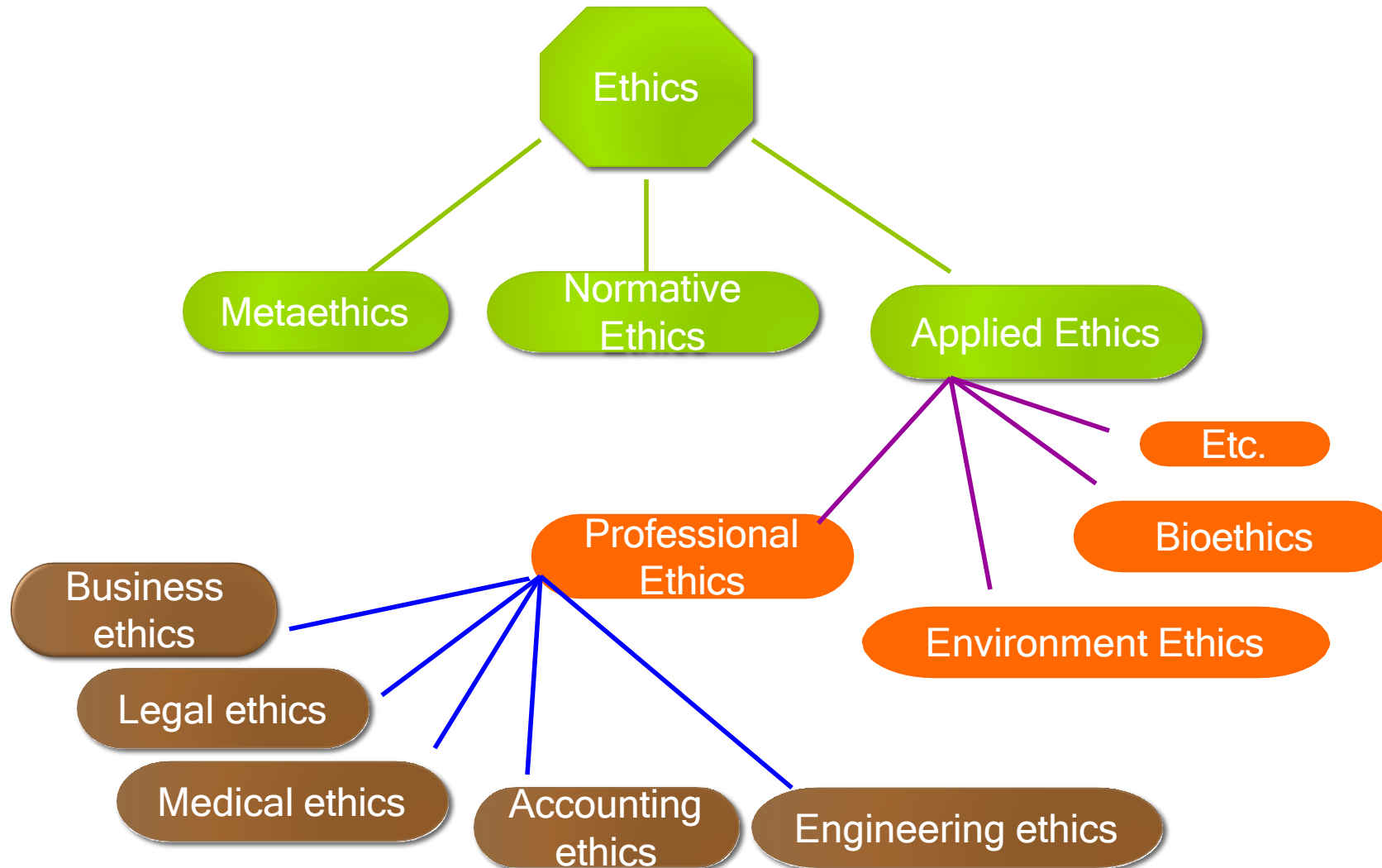
Is it ethical to remove safety parts to increase profit?

This is **applied ethics**.

Difference Table

Feature	Meta-Ethics	Normative Ethics	Applied Ethics
Main Question	What is right/wrong?	What should we do?	What should we do in this case?
Focus	Meaning of moral terms	Moral rules & principles	Real-world moral issues
Nature	Theoretical	Rule-based	Practical
Deals with	Truth of moral values	Standards of behavior	Specific situations
Example Question	“Is morality objective?”	“Is lying wrong?”	“Is it okay to lie to save a life?”
Example Area	Moral language, moral truth	Utilitarianism, duty, virtue	Medical, business, engineering ethics

What is Applied Ethics?



INTRODUCTION (SCENARIO)

A Real-Life Situation:

A software engineer finds a serious bug in a hospital system.
Fixing it delays the project and costs money.
Ignoring it may risk patient lives.

Example:

Wrong medicine dosage due to software error.

Questions:

- What should the engineer do?
- Is this a technical problem or an ethical problem?

ENGINEERING

Engineering:

- Application of science and mathematics
- To design, build, and solve real-world problems
- Serves society and improves life

Example:

Designing bridges, power plants, mobile apps, medical machines.

Questions:

- Is engineering only about machines?
- How does engineering help society?

PROFESSION

Profession:

- Requires special education
- Has social responsibility
- Has rules and standards

Example:

Doctors protect health, engineers protect safety.

Question:

- Why is engineering called a profession, not just a job?

A **job** is work done mainly for earning money. A **profession** like engineering requires **expert knowledge, ethical standards, societal responsibility, and formal recognition.** That combination is what makes engineering a profession, not just a job.

PROFESSIONALISM

Professionalism:

- Honesty
- Discipline
- Respect for time
- Responsibility

Example:

Engineer reports true test results even when boss pressures him.

Questions:

- Is technical skill alone enough?
- Why is honesty important?

1. Is technical skill alone enough?

Answer: No, technical skill alone is **not enough** to be a professional engineer.

Reason:

- Technical skill lets you **design and solve problems**, but **professionalism involves more than just technical ability**.
- Engineers interact with **clients, colleagues, and the public**, so **communication, teamwork, and ethical judgment** are equally important.
- Mistakes in judgment, ethics, or communication can cause **serious harm**, even if the technical work is perfect.

Example:

An engineer may know how to design a bridge technically, but if they **ignore safety codes to save money**, their skill alone cannot prevent accidents.

2. Why is honesty important?

Answer: Honesty is essential because **engineering decisions affect lives and society**.

Reason:

- Engineers must report **accurate data, risks, and results**.
- Hiding mistakes, exaggerating capabilities, or falsifying reports can lead to **disasters, loss of trust, or legal consequences**.
- Honesty builds **professional credibility and public trust**, which are critical for the engineering profession.

Example:

A civil engineer notices a defect in a building foundation. Being honest and reporting it prevents a potential collapse, even if it delays the project.

RESPONSIBILITIES OF ENGINEERS

Engineers are responsible to:

- Society (public safety)
- Employer (honest work)
- Client (quality service)
- Law (legal compliance)
- Profession (good image)

Example:

Refusing to approve unsafe building design.

Questions:

- Who suffers if an engineer ignores safety?
- Which responsibility is most important?

1. Who suffers if an engineer ignores safety?

If an engineer ignores safety, **multiple parties can suffer:**

- **Public / Users:** People using the product, building, or system are at risk of **injury or death**.
- **Clients / Employers:** Financial losses, and reputational damage.
- **Society:** Loss of trust in engineering and institutions, environmental damage.
- **The Engineer Themselves:** Legal action, loss of license, career damage, or moral guilt.

Example:

If a structural engineer ignores safety codes in a bridge design:

- People using the bridge could die or get injured.
- The construction company could face lawsuits.
- The engineer could lose their professional license.

2. Which responsibility is most important?

While engineers have many responsibilities (to clients, employers, society, environment), **the most important is protecting public safety and welfare.**

Reason:

- Lives are priceless; no technical success or profit justifies endangering humans.

Rana Plaza Collapse (Bangladesh, 2013)



It is April 2013 in Savar, near Dhaka. The eight-story **Rana Plaza** building houses several garment factories producing clothing for international brands. A day before the incident, **visible cracks** appear across multiple floors. Local engineers and inspectors inform the building owner that the structure is unsafe and should be **immediately evacuated**.

Banks and shops on the lower floors close after the warning. However, factory owners—under pressure to meet **tight export deadlines**—tell workers to return the next day. Many workers feel unsafe, but they are warned: *“If you don’t come, you will lose your job and wages.”*

On the morning of April 24, thousands of workers enter the building. Heavy generators are started (due to a power outage), causing intense vibration. Within hours, the building **collapses completely**, trapping workers under rubble.

Rescue operations last for days. The disaster sparks global outrage about **worker safety, engineering responsibility, and corporate ethics**.

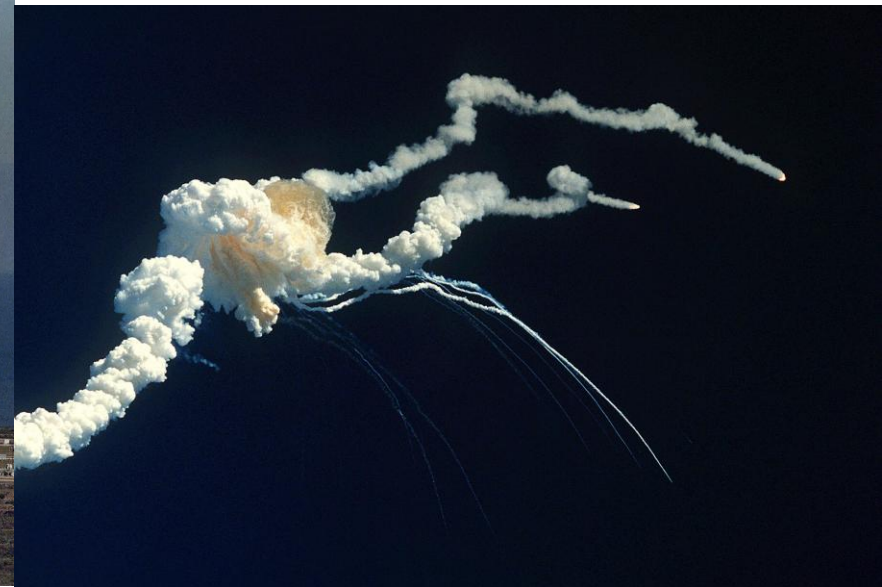
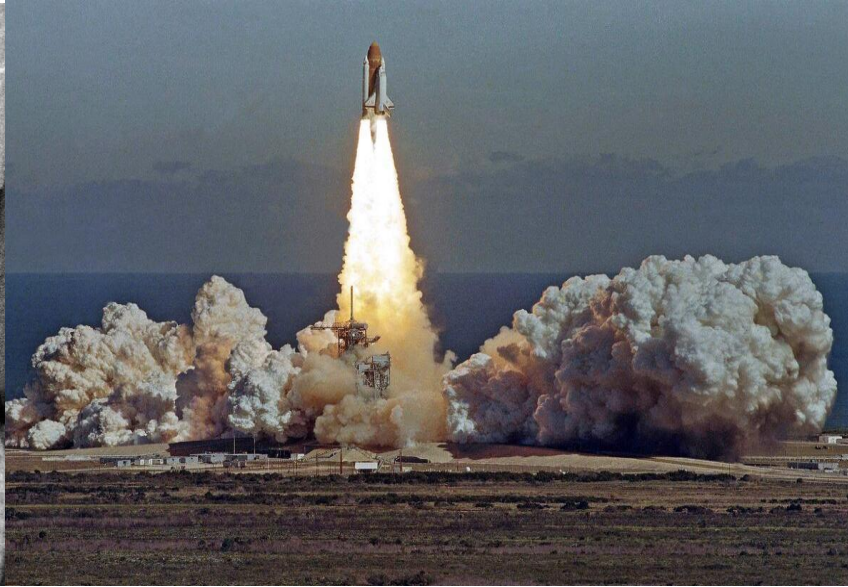
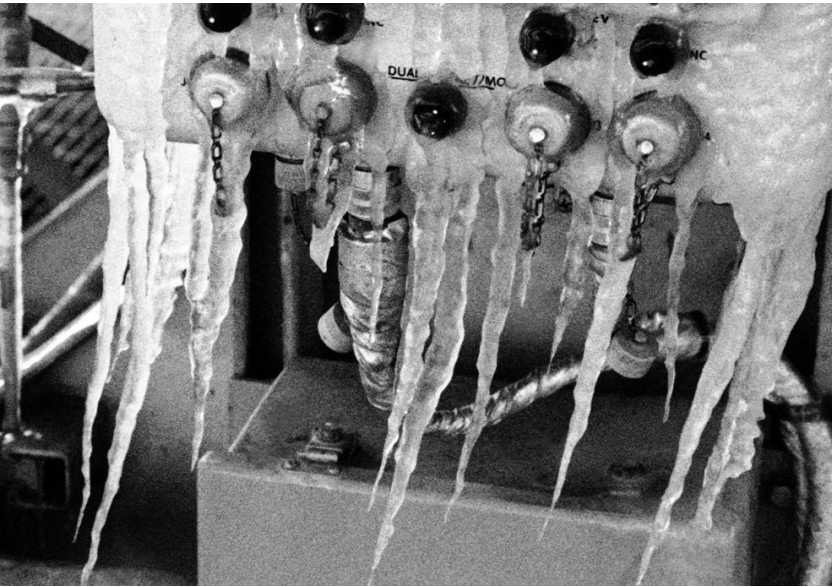
Ethical Dilemmas:

- Should engineers have taken stronger action to stop entry?
- Is economic pressure a valid excuse to ignore safety?
- What is the responsibility of global brands?

Role Play:

Engineer vs Factory Owner vs Worker vs Government
Inspector

Challenger Space Shuttle Disaster (USA, 1986)



You are an engineer at a contractor company working with NASA. The **Challenger shuttle** is scheduled to launch, but the weather is unusually **cold**. You know that a critical component—the **O-ring seals** in the rocket boosters—becomes brittle at low temperatures.

During a late-night meeting, you and your team strongly recommend **delaying the launch**. You present data showing potential failure risks. However, NASA managers are under **political and public pressure** to launch on time.

A tense discussion follows. Management questions your data and asks for proof that failure will definitely happen. Under pressure, senior executives override the engineers' concerns and approve the launch.

The next morning, the shuttle launches. Within 73 seconds, it **explodes**, killing all seven astronauts.

Ethical Dilemmas:

- Should engineers refuse to approve unsafe launches?
- How do you act when management ignores technical evidence?
- Is silence equal to responsibility?

Role Play:

Engineer vs NASA Manager vs Government Official vs Public

Uber Self-Driving Car Accident (2018)



You are part of a software engineering team developing **autonomous vehicles**. Your company is racing competitors to launch the first fully self-driving car. The system uses sensors (LiDAR, cameras) and AI algorithms to detect obstacles.

During testing, engineers notice that the system sometimes **misclassifies objects** (e.g., pedestrian vs shadow). Instead of delaying the project, management decides to continue testing on **public roads** to collect more data.

One night, a test vehicle is driving in autonomous mode. A pedestrian crosses the road pushing a bicycle. The system detects something but fails to classify it correctly in time. The safety driver inside the car is distracted.

The car does not brake in time and **hits the pedestrian**, causing death. Investigations reveal multiple failures: software design, safety monitoring, and testing decisions.

Ethical Dilemmas:

- Is it ethical to test incomplete AI systems on the public?
- Who is responsible for AI mistakes?
- Should engineers stop deployment if risks exist?

Role Play:

Software Engineer vs Company CEO vs Safety Driver vs Public Citizen

Volkswagen Emission Scandal (2015)



You are a software engineer at a major automobile company. The company promises customers that its diesel cars are **environmentally friendly**. However, during internal testing, engineers discover that the engines **cannot meet emission standards** without reducing performance.

Management faces a dilemma: redesign the engine (costly and time-consuming) or find a shortcut. Engineers are instructed to develop software that can **detect when the car is being tested** and temporarily reduce emissions only during testing.

In real-world driving, the cars emit pollutants far above legal limits.

Some engineers feel uncomfortable but continue working due to fear of losing their jobs. Years later, regulators discover the deception. The company faces massive fines, lawsuits, and loss of public trust.

Ethical Dilemmas:

- Should engineers follow unethical instructions?
- Is job security more important than honesty?
- How should whistleblowers act?

Role Play:

Software Engineer vs Company Executive vs Regulator vs Customer

“If you were designing a product, what trade-offs would you refuse to make?”

Role Play

Divide class into groups:

- Engineer
- Manager
- Client
- Public

Problem:

“Unsafe building design detected.”

Act out:

- Engineer explains
- Manager pressures
- Public demands safety

MORALS

Morals:

- Personal beliefs of right and wrong
- Learned from family, culture, religion
- Differ from person to person

Example:

“Lying is wrong”

“Helping others is good”

Questions:

- Do all people have same morals?
- Can morals conflict with job orders?

ETHICS

Ethics:

- Rules of right and wrong for a profession
- Same for all engineers
- Written in codes

Example:

Engineer must ensure safety of users.

Questions:

- Who decides professional ethics?
- Why do professions need ethics?

MORALS VS ETHICS

Morals: personal

Ethics: professional

Example:

Personal belief: “Don’t lie”

Professional rule: “Report test results honestly”

Question:

- Which is more important in job life: morals or ethics?

SCENARIO

Scenario:

A real-life situation where ethical decision is needed.

Examples:

- Data leak
- Bridge collapse
- Fake test report
- Software failure

Questions:

- Why do we use scenarios in ethics study?
- Is every problem a scenario?

ETHICS IN ENGINEERING

Ethics in Engineering:

- Applying ethics in engineering work
- Protecting human life
- Being honest
- Refusing unsafe practices

Example:

Stopping release of unsafe software.

Questions:

- Is profit more important than safety?
- Should engineer lose job to protect people?

CIVIC VIRTUE

Civic Virtue:

- Duty to society
- Obeying law
- Working for public good
- Avoiding corruption

Example:

Engineer refusing bribe to approve unsafe building.

Questions:

- Is corruption a private or social problem?
- How does civic virtue help nation?

VALUING TIME

Valuing Time:

- Being punctual
- Meeting deadlines
- Respecting others' time
- Efficient work

Example:

Engineer completes project on time.

Questions:

- Is being late an ethical problem?
- Who suffers when time is wasted?

TIME AS AN ETHICAL ISSUE

Time misuse causes:

- Project delay
- Financial loss
- Poor service

Example:

Signing attendance and leaving office.

Questions:

- Is time theft equal to money theft?
- Should company punish time misuse?

ETHICAL REASONING SKILL

Ethical reasoning = ability to decide what is right.

4 Steps:

1. Identify problem
2. Identify stakeholders
3. List options
4. Choose ethical action

Example:

Company asks to hide defect.

Question:

- Why should engineers learn ethical reasoning?

ETHICAL REASONING (EXAMPLE)

Case: Hide test failure

1.Problem: Safety risk

2.Stakeholders: Users, company, engineer

3.Options: Hide or report

4.Best: Report

Question:

- What if company threatens to fire you?

CODE OF ETHICS

◆ What is a Code of Ethics?

A **Code of Ethics** is a set of principles that guide professionals to behave **honestly, responsibly, and fairly** in their work.

Code of Ethics:

- Written rules
- Guides behavior
- Protects public interest

Example: 1: Honesty and Integrity Rule: Be truthful and do not mislead others.

A software engineer finds a bug in a banking app before release. Instead of hiding it to meet the deadline, they report it immediately—even if it delays the project.

 **Ethical Point: Honesty is more important than deadlines.**

Question:

- Why not depend only on personal morals?

PURPOSE OF CODE OF ETHICS

- Ensures safety
- Builds trust
- Prevents corruption
- Protects profession image

Example:

Public trusts engineers who follow code.

Question:

- What happens when engineers break code of ethics?

IMPORTANT ETHICAL PRINCIPLES

- Safety first
- Honest reporting
- No bribe
- Respect colleagues
- Maintain competence

Example:

Rejecting fake certificate.

Question:

- Which principle is most important and why?

CASE STUDY

Engineer asked to approve unsafe building for bribe.

Questions:

- Who is affected?
- What is ethical action?
- Which rule is broken?

KEY TAKEAWAYS

- Engineering serves society
- Professionalism = honesty + responsibility
- Morals = personal, Ethics = professional
- Civic virtue = social duty
- Code of ethics guides engineers

Question:

- What kind of engineer do you want to be?

REFLECTION

Think:

- Would you hide a dangerous bug?
- Is money more important than safety?

Question:

- Write one ethical rule you will follow as an engineer.



NATIONAL SOCIETY OF
PROFESSIONAL ENGINEERS

Code of Ethics for Engineers

Preamble

Engineering is an important and learned profession. As members of the profession, engineers are expected to exhibit the highest standards of honesty and integrity. Engineering has a direct and vital impact on the quality of life for all people. Accordingly, the services provided by engineers require honesty, impartiality, fairness, and equity, and must be dedicated to the protection of the public health, safety, and welfare. Engineers must perform under a standard of professional behavior that requires adherence to the highest principles of ethical conduct.

I. Fundamental Canons

Engineers, in the fulfillment of their professional duties, shall:

1. Hold paramount the safety, health, and welfare of the public.
2. Perform services only in areas of their competence.
3. Issue public statements only in an objective and truthful manner.
4. Act for each employer or client as faithful agents or trustees.
5. Avoid deceptive acts.
6. Conduct themselves honorably, responsibly, ethically, and lawfully so as to enhance the honor, reputation, and lawfulness of the profession.

II. Rules of Practice

1. Engineers shall hold paramount the safety, health, and welfare of the public.

- a. If engineers' judgment is overruled under circumstances that endanger life or property, they shall advise their employer or client and such other authority as may be appropriate.
- b. Engineers shall acknowledge only those engineering documents that are in conformity with applicable standards.
- c. Engineers shall not reveal facts, data, or information without the prior consent of the client or employer except as authorized or required by law or this Code.
- d. Engineers shall not permit the use of their name or associate in business ventures with any person or firm that they believe is engaged in fraudulent or dishonest enterprise.
- e. Engineers shall not aid or abet the unlawful practice of engineering by a person or firm.
- f. Engineers having knowledge of any alleged violation of this Code shall report thereon to appropriate professional bodies and, when relevant, also to public authorities, and cooperate with the proper authorities in furnishing such information or assistance as may be required.

2. Engineers shall perform services only in the areas of their competence.

- a. Engineers shall undertake assignments only when qualified by education or experience in the specific technical fields involved.
- b. Engineers shall not affix their signatures to any plans or documents dealing with subject matter in which

they lack competence, nor to any plan or document not prepared under their direction and control.

- a. Engineers shall accept assignments and assume responsibility for coordination of an entire project and sign and seal the engineering documents for the entire project, provided that each technical segment is signed and sealed only by the qualified engineers who prepared the segment.

3. Engineers shall issue public statements only in an objective and truthful manner.

- a. Engineers shall be objective and truthful in professional reports, statements, or testimony. They shall include all relevant and pertinent information in such reports, statements, or testimony, which should bear the date indicating when it was current.
- b. Engineers may express publicly technical opinions that are founded upon knowledge of the facts and competent in the subject matter.
- c. Engineers shall issue no statements, opinions, or arguments on technical matters that are inspired or paid for by interested parties, unless they have disclosed their comments by explicitly identifying the interested parties in whose behalf they are speaking, and by revealing the existence of any interest the engineers may have in the matter.

4. Engineers shall act for each employer or client as faithful agents or trustees.

- a. Engineers shall disclose all known or potential conflicts of interest that could influence or appear to influence their judgment or the quality of their services.
- b. Engineers shall not accept compensation, financial or otherwise, from more than one party for services on the same project, or for services pertaining to the same project, unless the circumstances are fully disclosed and agreed to by all interested parties.
- c. Engineers shall not solicit or accept financial or other valuable consideration, directly or indirectly, from outside agents in connection with the work for which they are responsible.
- d. Engineers in public service as reviewers, advisors, or employees of a governmental or quasi-governmental body or department shall not participate in decisions with respect to services solicited or provided by them or their organizations in private or public engineering practice.
- e. Engineers shall not solicit or accept a contract from a governmental body on which a principal or officer of their organization serves as a member.

5. Engineers shall avoid deceptive acts.

- a. Engineers shall not identify their qualifications or permit misrepresentation of their or their associates' qualifications. They shall not misrepresent or exaggerate their responsibility in or for the subject matter of prior assignments. Brochures or other presentations must not

be the solicitation of employment shall not misrepresent pertinent facts concerning employers, employees, associates, joint ventures, or past accomplishments.

- b. Engineers shall not offer, give, solicit, or receive, either directly or indirectly, any contribution to influence the award of a contract by public authority, or which may be reasonably construed by the public as having the effect or intent of influencing the awarding of a contract. They shall not offer any gift or other valuable consideration in order to secure work. They shall not pay a commission, percentage, or brokerage fee in order to secure work, except to a bona fide employee or bona fide established commercial or marketing agencies retained by them.

III. Professional Obligations

1. Engineers shall be guided in all their relations by the highest standards of honesty and integrity.

- a. Engineers shall acknowledge their errors and shall not distort or alter the facts.
- b. Engineers shall advise their clients or employers when they believe a project will not be successful.
- c. Engineers shall not accept outside employment to the detriment of their regular work or interest. Before accepting any outside engineering employment, they will notify their employers.
- d. Engineers shall not attempt to attract an engineer from another employer by false or misleading pretenses.
- e. Engineers shall not promote their own interest at the expense of the dignity and integrity of the profession.
- f. Engineers shall treat all persons with dignity, respect, fairness, and without discrimination.

2. Engineers shall at all times strive to serve the public interest.

- a. Engineers are encouraged to participate in their efforts toward guidance for society and work for the advancement of the safety, health, and well-being of their community.
- b. Engineers shall not complete, sign, or seal plans and/or specifications that are not in conformity with established engineering standards, if the client or employer insists on such unprofessional conduct, they shall notify the proper authorities and withdraw from further service on the project.
- c. Engineers are encouraged to extend public knowledge and appreciation of engineering and its achievements.
- d. Engineers are encouraged to adhere to the principles of sustainable development in order to protect the environment for future generations.
- e. Engineers shall continue their professional development throughout their careers and should keep current in their specialty fields by engaging in professional practice, participating in continuing education courses, reading in the technical literature, and attending professional meetings and seminars.

The **NSPE Code of Ethics for Engineers** is the ethical framework published by the [National Society of Professional Engineers \(NSPE\)](#). It establishes the professional responsibilities engineers owe to the public, clients, employers, and the engineering profession. The full, official code is available here: [NSPE Code of Ethics for Engineers](#).

The Six Fundamental Canons

Engineers, in the fulfillment of their professional duties, shall:

- 1. Hold paramount the safety, health, and welfare of the public.**
- 2. Perform services only in the areas of their competence.**
- 3. Issue public statements only in an objective and truthful manner.**
- 4. Act for each employer or client as faithful agents or trustees.**
- 5. Avoid deceptive acts.**
- 6. Conduct themselves honorably, responsibly, ethically, and lawfully so as to enhance the honor, reputation, and usefulness of the profession.**

Structure of the Code

The NSPE Code consists of three main sections:

- **Fundamental Canons** – The six overarching ethical principles.
- **Rules of Practice** – Specific requirements for professional conduct, such as protecting public safety, practicing within one's competence, avoiding conflicts of interest, and maintaining objectivity.
- **Professional Obligations** – Guidance on honesty, integrity, confidentiality, professional development, giving proper credit, and supporting the engineering profession.

Why It Matters

The NSPE Code serves as a guide for ethical decision-making and emphasizes that an engineer's foremost responsibility is protecting the public. It is widely used in engineering education, professional licensure, and workplace ethics, although state licensing boards may have their own legally enforceable codes and regulations.

1. Hold paramount the safety, health, and welfare of the public

Public safety must always come first—even before company profit or deadlines.

Real-life example:

An engineer discovers that a bridge design has a structural weakness. The company wants to continue construction to save time and money.

👉 The engineer must report and stop the project.

✓ Saving lives is more important than profit.

2. Perform services only in areas of competence

Engineers should only work in fields where they have proper knowledge and skills.

Real-life example:

A software engineer is asked to design a chemical plant system.

👉 They should refuse or consult a chemical engineer.

✓ Working outside expertise can cause serious harm.

3. Issue public statements only in an objective and truthful manner

Always be honest and avoid misleading information.

Real-life example:

An engineer is asked to hide test results showing a product defect.

👉 They must report the truth.

✓ False information can lead to accidents and legal issues.

4. Act for each employer or client as faithful agents or trustees

Be loyal and protect your client's or employer's interests (without breaking ethics).

Real-life example:

An engineer working for Company A secretly shares design details with Company B.

👉 This is unethical.

✓ Confidentiality must be maintained.

5. Avoid deceptive acts

Do not cheat, lie, or engage in corruption.

Real-life example:

An engineer accepts a bribe to approve a low-quality building.

👉 This violates ethics.

✓ Deception can cause disasters and legal punishment.

6. Conduct themselves honorably, responsibly, ethically, and lawfully

Maintain professionalism in all actions and respect laws and ethical standards.

Real-life example:

An engineer helps junior team members learn and follows all safety laws even under pressure.

✓ Builds trust and improves the profession's reputation.

IEEE Code of Ethics

We, the members of the IEEE, in recognition of the importance of our technologies in affecting the quality of life throughout the world, and in accepting a personal obligation to our profession, its members, and the communities we serve, do hereby commit ourselves to the highest ethical and professional conduct and agree:

I. To uphold the highest standards of integrity, responsible behavior, and ethical conduct in professional activities.

1. to hold paramount, the safety, health, and welfare of the public, to strive to comply with ethical design and sustainable development practices, to protect the privacy of others, and to disclose promptly factors that might endanger the public or the environment;
2. to improve the understanding by individuals and society of the capabilities and societal implications of conventional and emerging technologies, including intelligent systems;
3. to avoid real or perceived conflicts of interest whenever possible, and to disclose them to affected parties when they do exist;
4. to avoid unlawful conduct in professional activities, and to reject bribery in all its forms;
5. to seek, accept, and offer honest criticism of technical work, to acknowledge and correct errors, to be honest, and realistic in stating claims or estimates based on available data, and to credit properly the contributions of others;
6. to maintain and improve our technical competence and to undertake technological tasks for others only if qualified by training or experience, or after full disclosure of pertinent limitations;

II. To treat all persons fairly and with respect, to avoid harassment or discrimination, and to avoid injuring others.

7. to treat all persons fairly and with respect, and to not engage in discrimination based on characteristics such as race, religion, gender, disability, age, national origin, sexual orientation, gender identity, or gender expression;
8. to not engage in harassment of any kind, including sexual harassment or bullying behavior;
9. to avoid injuring others, their property, reputation, or employment by false or malicious actions, rumors, or any other verbal or physical abuses;

III. To strive to ensure this code is upheld by colleagues and co-workers.

10. to support colleagues and co-workers in following this code of ethics, to strive to ensure the code is upheld, and to not retaliate against individuals reporting a violation.

**Adopted by the
IEEE Board of Directors
June 2020**

IEEE Code of Ethics (Simplified)

IEEE members agree to uphold the following key principles:

1. Prioritize safety, health, and public welfare

Always consider how your work affects society and human well-being.

Example:

An engineer stops releasing a faulty medical device that could harm patients.

2. Avoid conflicts of interest and disclose them

Do not let personal gain influence professional decisions.

Example:

If you are working for two competing companies, you must disclose it.

3. Be honest and realistic in claims and estimates

Do not exaggerate results or hide risks.

Example:

A software engineer accurately reports system limitations instead of overpromising.

4. Reject bribery and corruption

Never accept or offer bribes.

Example:

Refusing money to approve a low-quality engineering project.

5. Improve understanding of technology and its consequences

Help society understand both benefits and risks of technology.

Example:

Explaining AI risks like bias or privacy concerns to users.

6. Maintain and improve technical competence

Keep learning and stay updated in your field.

Example:

Learning new cybersecurity techniques to prevent attacks.

7. Seek, accept, and offer honest criticism

Encourage feedback and correct mistakes.

Example:

Accepting code review feedback and improving your design.

8. Treat all people fairly (no discrimination)

Avoid bias based on gender, race, religion, etc.

Example:

Ensuring an AI hiring system treats all candidates equally.

9. Avoid harming others (reputation, career, property)

Do not damage others through false or malicious actions.

Example:

Not spreading false information about a colleague.

10. Support colleagues and ethical practices

Help others follow ethical behavior.

Example:

Reporting unethical actions and supporting whistleblowers

Difference from NSPE (Quick Insight)

- **NSPE:** Focuses more on *professional engineering practice* (construction, safety, licensing)
- **IEEE:** Focuses more on *technology, computing, and global impact*

NSPE vs IEEE Code of Ethics (Key Differences)

Aspect	NSPE Code of Ethics	IEEE Code of Ethics
Organization	National Society of Professional Engineers	Institute of Electrical and Electronics Engineers
Primary Focus	Public safety, engineering practice (civil, mechanical, etc.)	Technology ethics (AI, software, electronics, computing)
Scope	Mostly U.S.-centric professional engineering practice	Global and technology-oriented
Core Principle	“Hold paramount public safety” (very strong emphasis)	Balanced principles: safety, honesty, fairness, competence
Structure	6 Canons + Rules of Practice + Professional Obligations	10 concise ethical principles
Legal Connection	Closely tied to licensing (PE) and legal responsibility	Not legally binding; more professional guidance
Technical Focus	Less focus on modern tech issues	Strong focus on AI, data, cybersecurity, emerging tech
Conflict of Interest	Strong emphasis on loyalty to employer/client	Emphasis on disclosure and transparency
Honesty & Reporting	Focus on truthful public statements	Broader: honesty in claims, data, research, communication
Professional Development	Encouraged but less detailed	Strong emphasis on continuous learning
Social Impact	Focus on safety and welfare	Includes fairness, non-discrimination, societal impact

🤖 Case Study 1: AI Ethics — “The Predictive Policing System”

A city deploys an AI system called **SafeCity AI** to predict crime hotspots and allocate police patrols. The model is trained on **historical crime data** from the past 10 years.

Ayesha, a data scientist on the team, notices:

- The model **over-predicts crime** in certain low-income neighborhoods.
- Police presence increases there, leading to **more arrests**, which feed back into the dataset (a **feedback loop**).
- Independent testing shows **higher false positives** for minority groups.
- City officials say, “Crime rates are dropping—don’t change the model before elections.”

Ayesha also learns the system’s decisions are **not explainable** to the public.



Questions

1. **Issue Identification (L2):** List at least 4 ethical issues.
2. **Standards Mapping (L3):** Which IEEE principles are at risk?
3. **Root Cause (L4):** Explain the feedback loop problem.
4. **Decision (L5):** What should Ayesha do first? Why?
5. **Design (L6):** Propose 3 technical fixes (e.g., fairness metrics, re-sampling, explainability).
6. **Policy (L5):** What governance measures should the city adopt?



Case Study 2: Cybersecurity — “The Silent Data Breach”

A fintech company, **PayLink**, stores sensitive customer data (IDs, transactions).

Rafi, a security engineer, detects unusual outbound traffic and confirms a **data breach** affecting ~200,000 users.

Findings:

- The breach exploited an **unpatched vulnerability** flagged 3 months ago.
- Management delayed patching due to **system downtime concerns**.
- Legal team advises: “Delay disclosure to avoid panic and reputational damage.”
- Attackers may have accessed **financial and identity data**.

Rafi is told to “monitor quietly” while leadership decides.



Questions

1. **Issue Identification (L2):** What went wrong operationally and ethically?
2. **Standards Mapping (L3):** Which IEEE principles apply (e.g., honesty, avoiding harm)?
3. **Immediate Action (L5):** What should Rafi do in the next 24 hours?
4. **Risk Analysis (L4):** Consequences of delayed disclosure?
5. **Compliance (L4):** What legal/industry obligations might apply (e.g., breach notification)?
6. **Prevention (L6):** Propose a 5-step incident response + patch management plan.



Case Study 3: Software Ethics — “The Rushed Release”

A startup, **MediTrack**, builds a mobile app for **medication reminders** and dosage tracking. **Nabila**, the lead developer, is preparing a major release.

Before launch:

- QA reports a **critical bug**: under certain conditions, the app shows the **wrong dosage**.
- Marketing has already announced the launch date; investors expect release.
- The CEO says: “It affects a small number of users—fix it in the next update.”
- There’s **no clear disclaimer** about potential inaccuracies.

Nabila must decide whether to proceed.

Questions

1. **Issue Identification (L2)**: What are the ethical concerns?
2. **Standards Mapping (L3)**: Which IEEE principles are violated if released?
3. **Decision (L5)**: Should Nabila release, delay, or partially roll out? Justify.
4. **Impact (L4)**: Potential harm to users and company?
5. **Mitigation (L6)**: Suggest release strategies (e.g., feature flags, phased rollout, kill switch).
6. **Professional Duty (L5)**: How should Nabila communicate with leadership?